Courier Management System

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Courier Management System (CMS)

1. INTRODUCTION

1.1 Project Overview :

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems.

These issues lead to delayed updates, poor customer visibility, and operational inefficiencies.

The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

Purpose

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability.

Problem Statement

In today's fast-growing e-commerce and logistics industry, courier companies handle a large number of parcel deliveries every day.

Many small and medium courier services still rely on manual record-keeping or outdated systems to manage bookings, track shipments, assign delivery agents, and update delivery status. This leads to errors, data loss, delays, lack of transparency, and poor customer satisfaction.

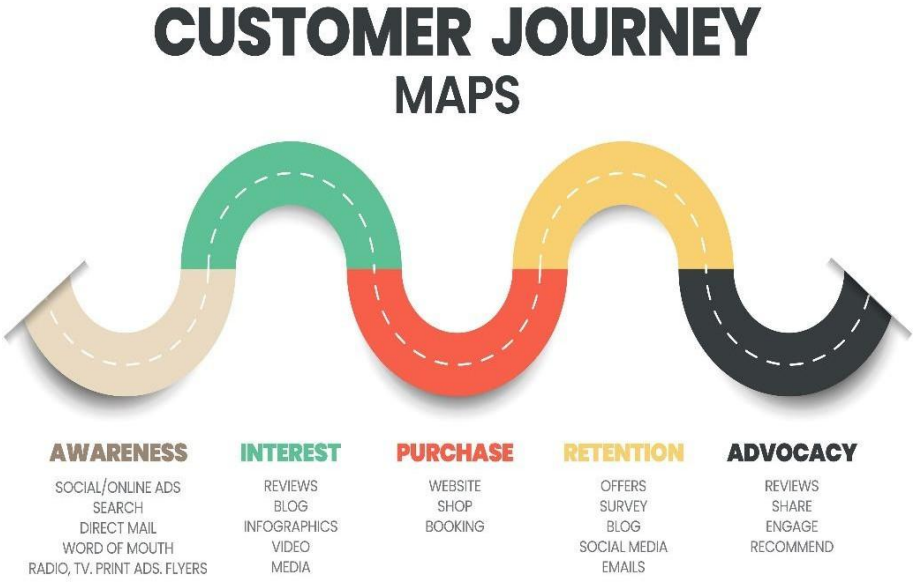
Customers often face difficulties in tracking their parcels in real time. Delivery staff struggle with inefficient route planning and manual updates. Administrators find it hard to monitor parcel status, generate reports, and manage customer records efficiently.

REQUIREMENT ANALYSIS PHASE

Customer Journey Map

Stages:

Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting.



Solution Requirements

Functional Requirements

1. Requirement
2. Customer management
3. Shipment booking
4. Delivery agent assignment
5. Real-time status updates
6. Invoice generation
7. Reports & dashboards

Non-Functional Requirements

1. NFR Description

2. Usability Easy Lightning interface
3. Security Role-based access control

4. Reliability Error-free automation

5. Performance Fast tracking updates

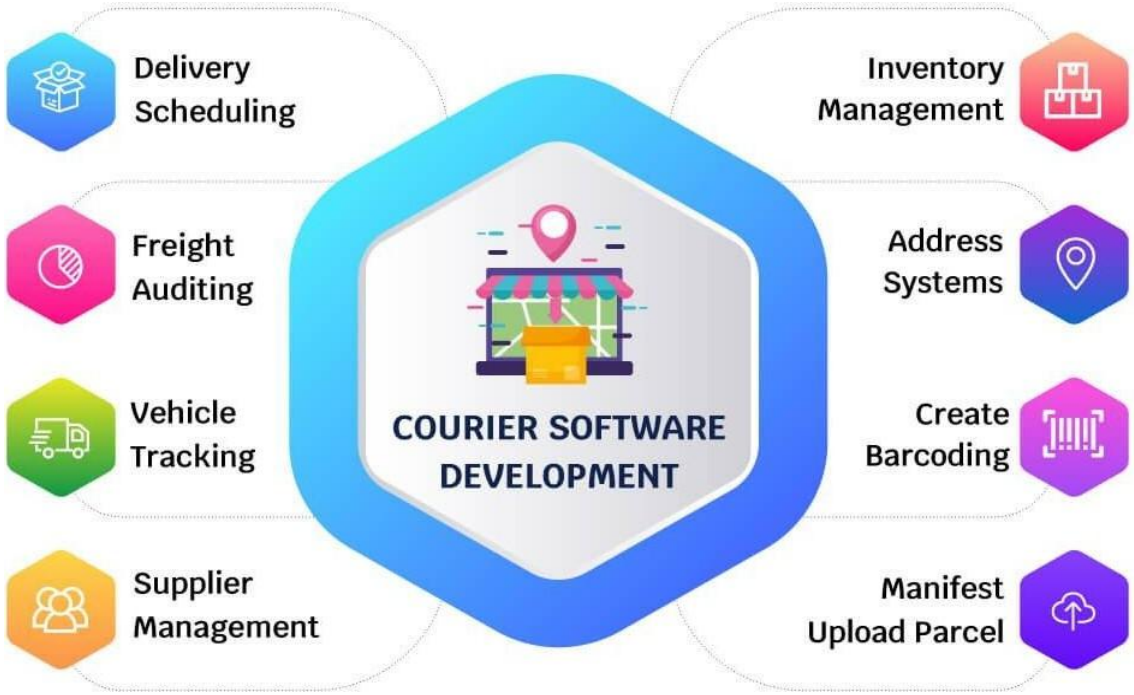
6. Availability 24/7 cloud access

7. Scalability Supports growing branches

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery
Update → Invoice → Reports



Technology Stack

1. Layer Technology
2. UI Salesforce Lightning
3. Logic Salesforce Flows
4. Database Salesforce Objects
5. Automation Validation Rules, Flows
6. Security Profiles, Roles
7. Reporting Reports & Dashboards

Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

Salesforce

Salesforce is a leading cloud-based software company that provides Customer Relationship Management (CRM) solutions. It helps businesses manage their sales, customer service, marketing, and operations in one centralized platform.

It was founded in 1999 by Marc Benioff and is headquartered in San Francisco.

What is Salesforce?

Salesforce is a cloud-based Customer Relationship Management (CRM) platform that helps businesses manage their customers, sales, service, and marketing activities in one system.

Instead of installing software on a computer, Salesforce works online (in the cloud), so companies can access it from anywhere using the internet.

PROJECT PLANNING & SCHEDULING

Project Planning

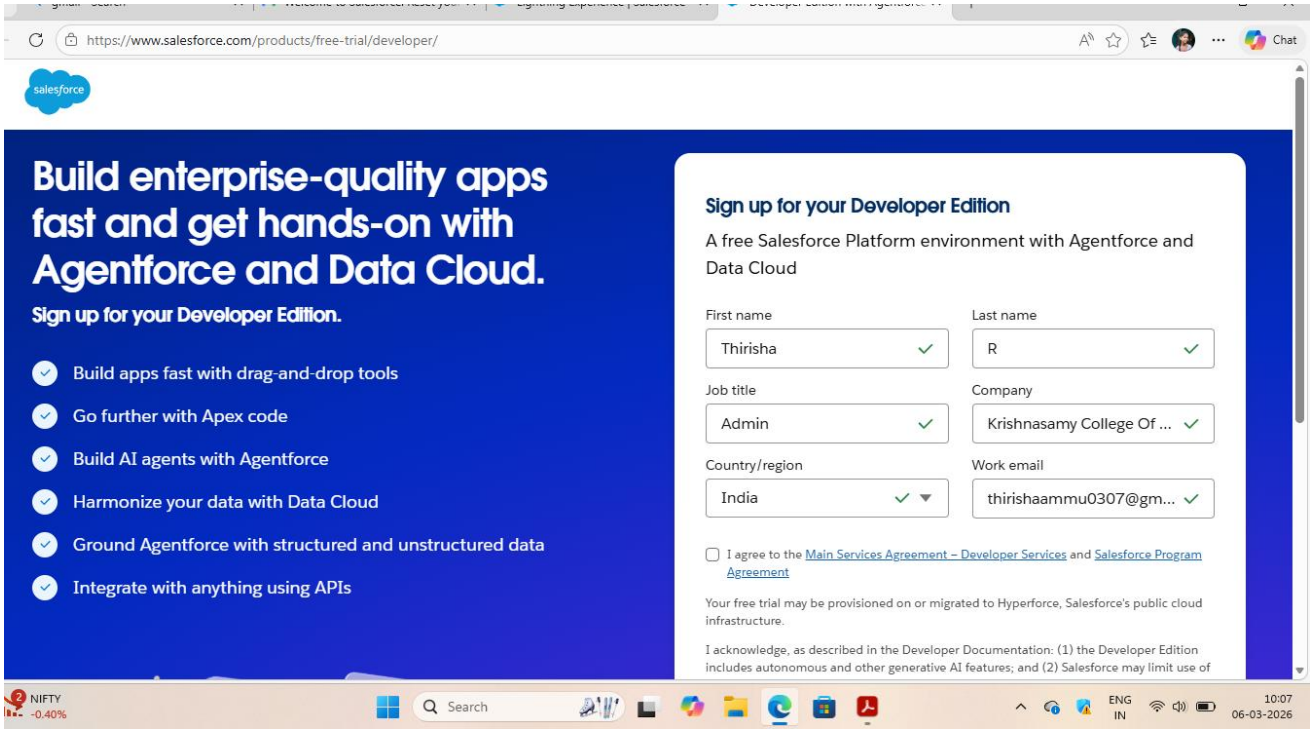
- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Project Development Phase:

1. Real-time Salesforce project implementation, Designing custom objects and relationships
2. Creating page layouts & record types , Writing validation rules
3. Building automation with flows, Developing reports & dashboards
4. Setting up security with profiles, permission sets, and sharing

End-to-end application testing

Salesforce Account



Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).
Custom Objects Created:

1. Customer__c – stores customer personal and contact information

2. Shipment__c – manages parcel details, source, destination, and tracking information

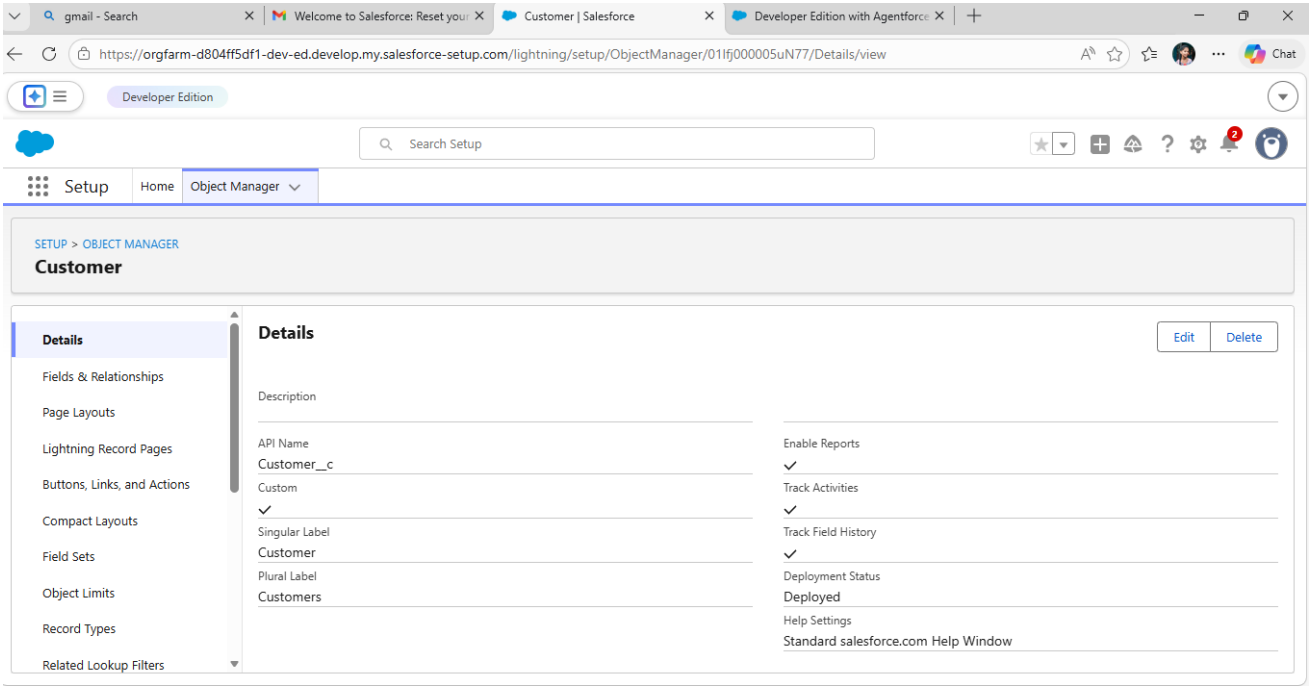
3. Delivery_Agent__c – maintains delivery agent profiles and branch details

4. Branch__c – represents courier branch offices

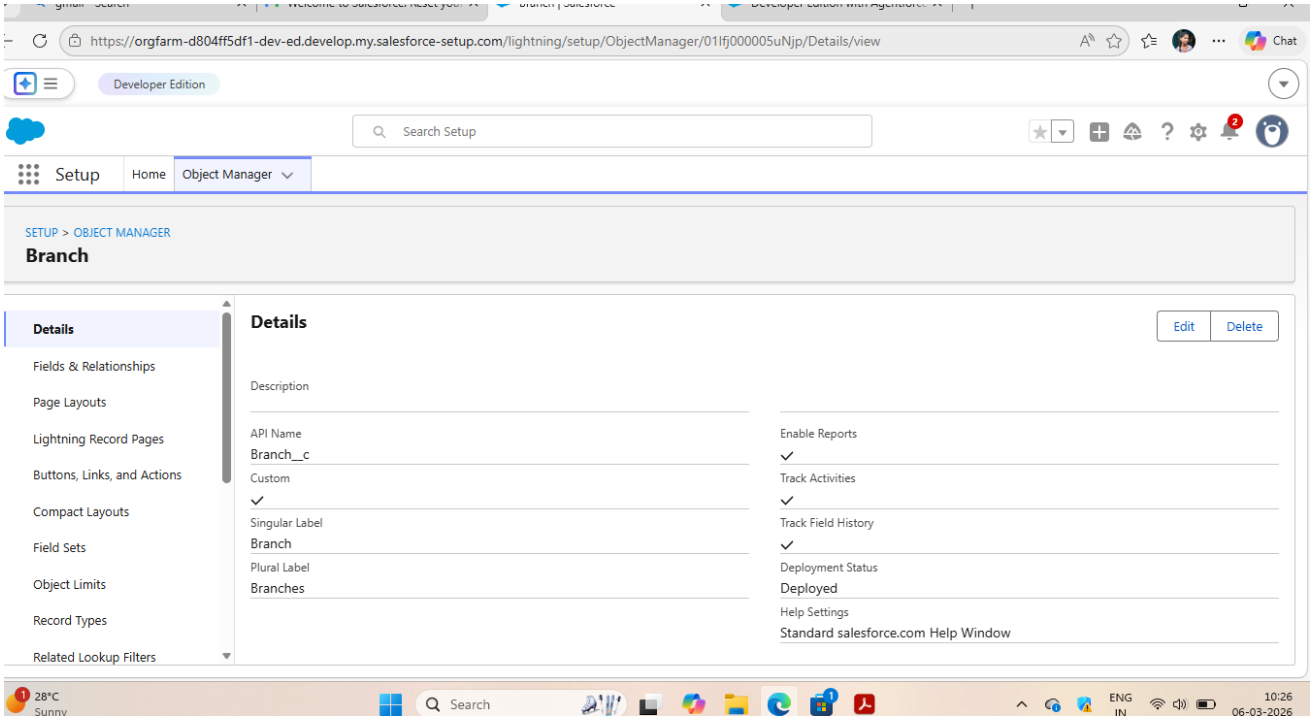
5 Delivery_Update__c – stores status updates for each shipment

6. Invoice__c – manages billing details for delivered shipments

Customer Object :



Branch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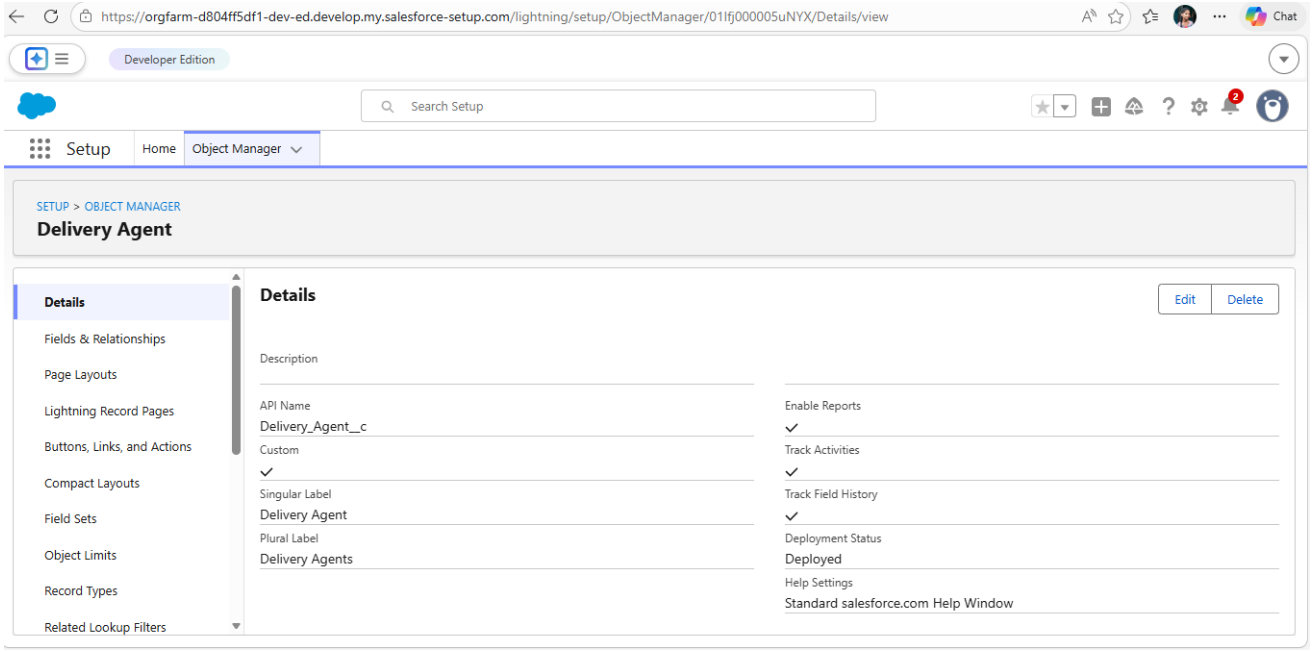
Object Manager
SETUP > OBJECT MANAGER
Branch
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Details
Edit
Delete
Description
API Name
Branch__c
Custom
Singular Label
Branch
Plural Label
Branches
Enable Reports
Track Activities
Track Field History
Deployment Status
Deployed
Help Settings
Standard salesforce.com Help Window

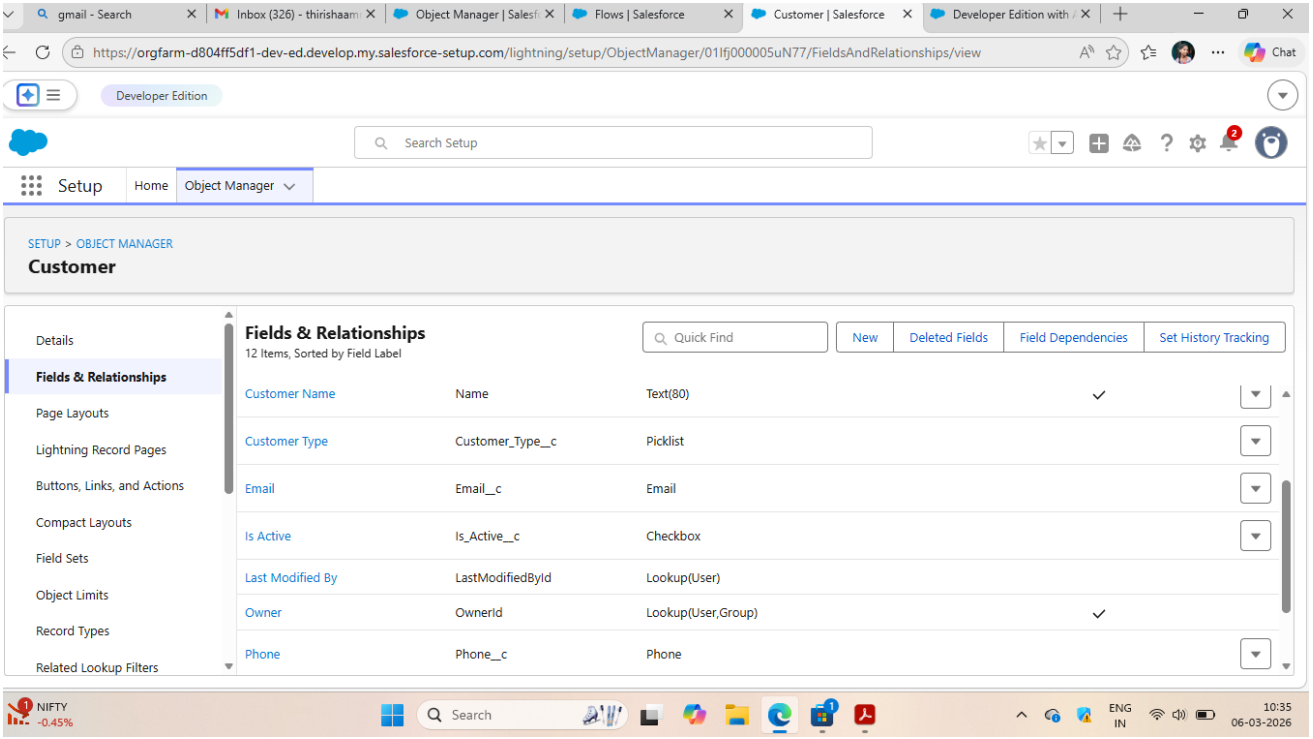
Delivery Agent



Fields & Relationships

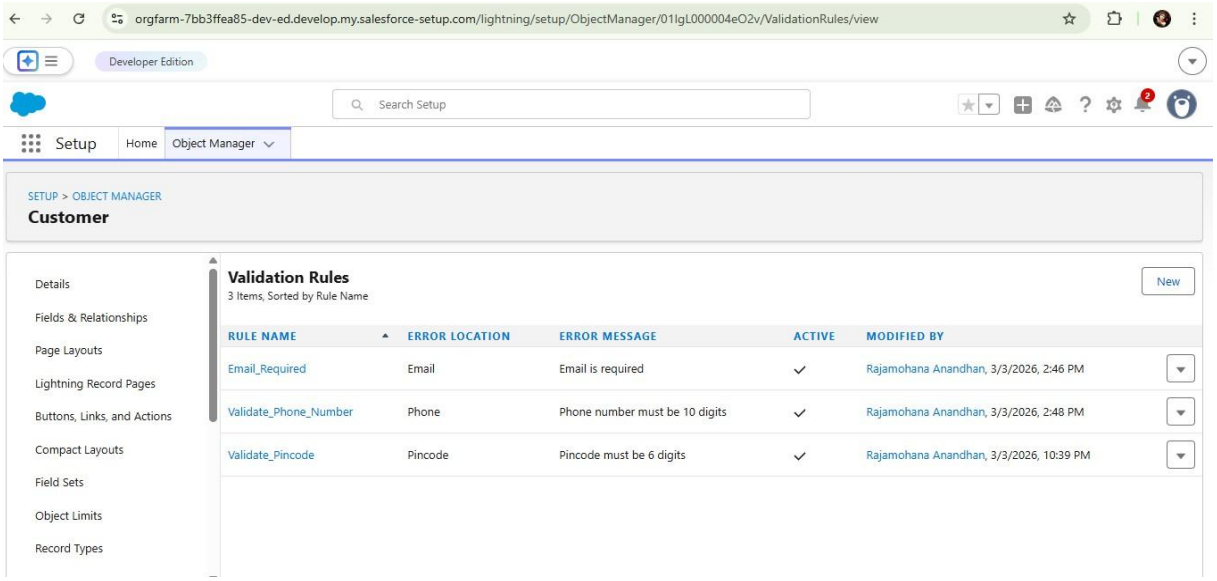
What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.



Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.



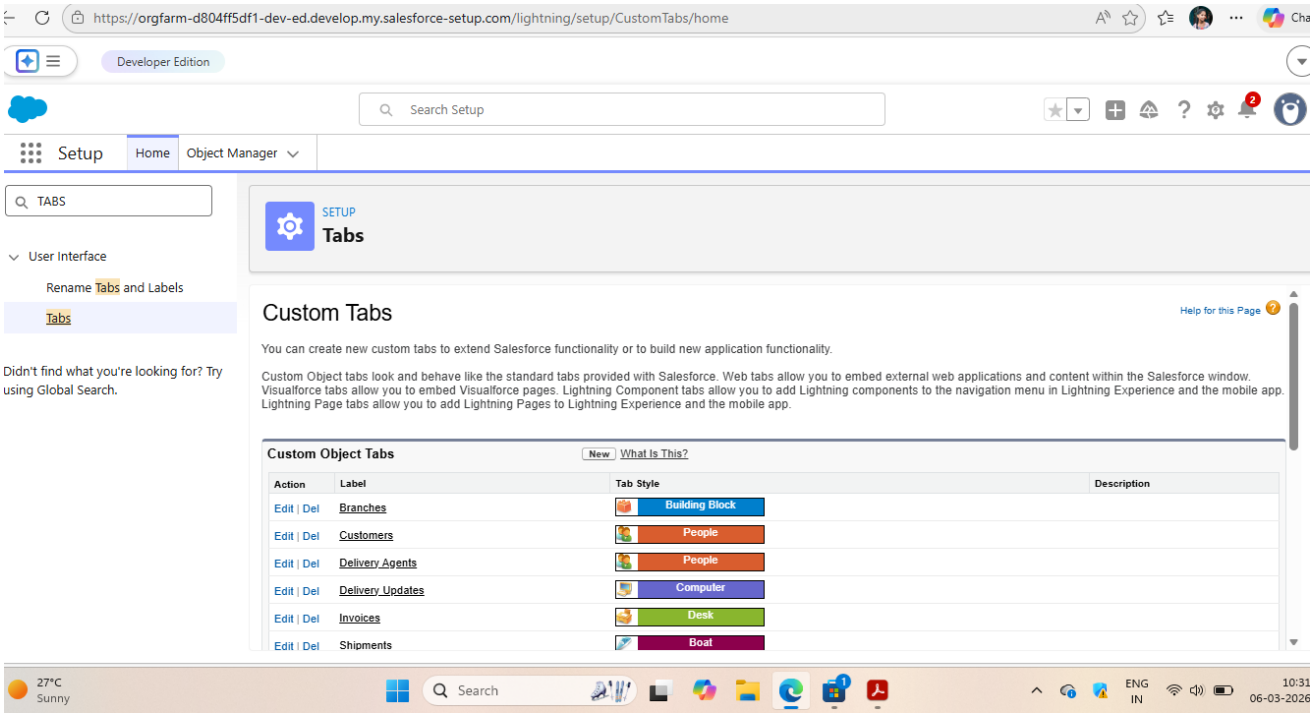
Tabs

What are Tabs in Salesforce?

Tabs are the navigation items at the top of the Salesforce page that allow users to quickly open objects, apps, or web pages

Example: Accounts, Contacts, Leads, Reports, Dashboard etc.

Tab Creation

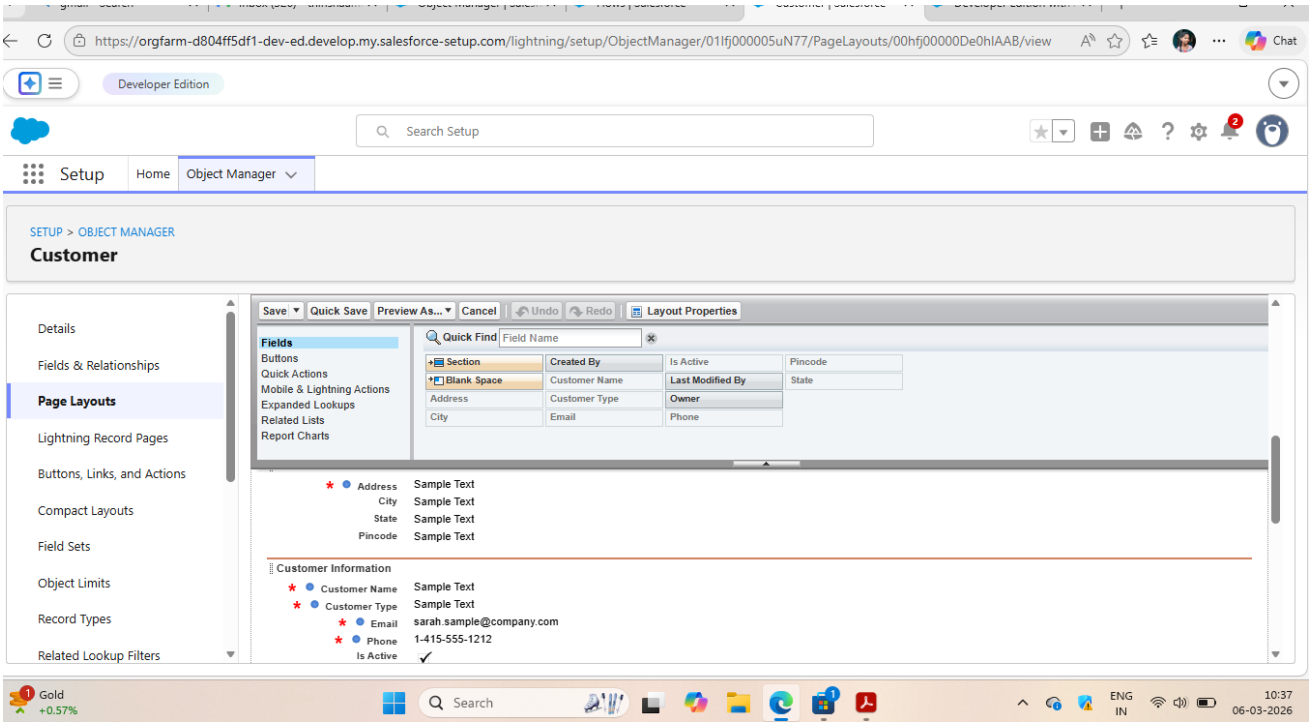


Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.

They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.

By customizing page layouts, we ensure that users can easily enter, update, and view the required data.



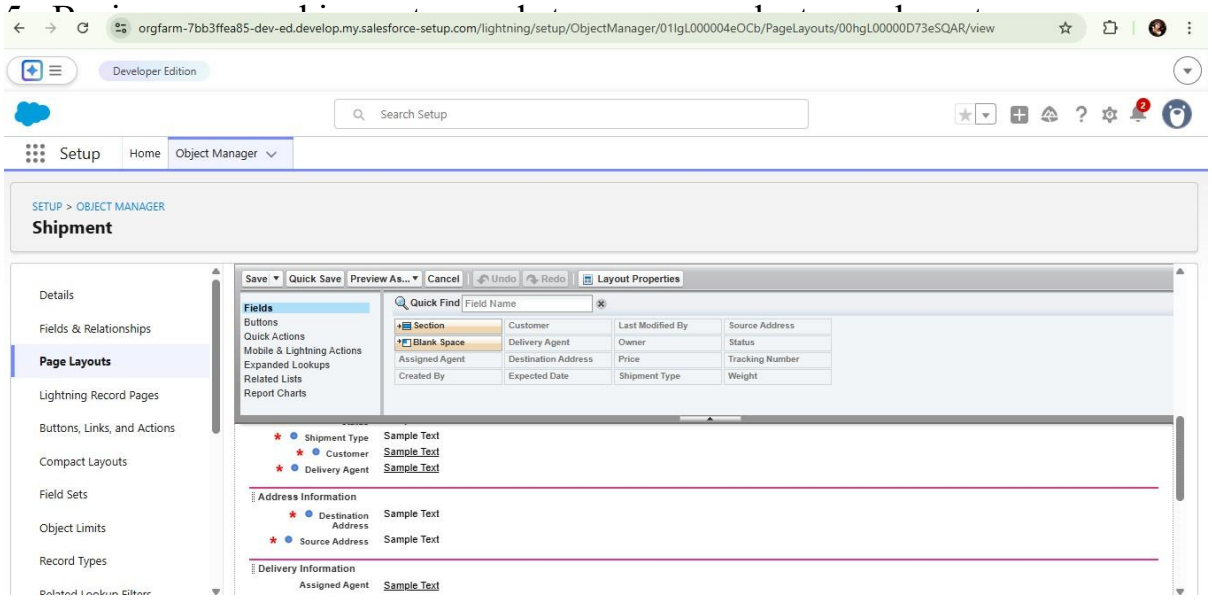
Shipment in Page Layout

1. Shipment is used to track the movement of products from the warehouse to the customer.

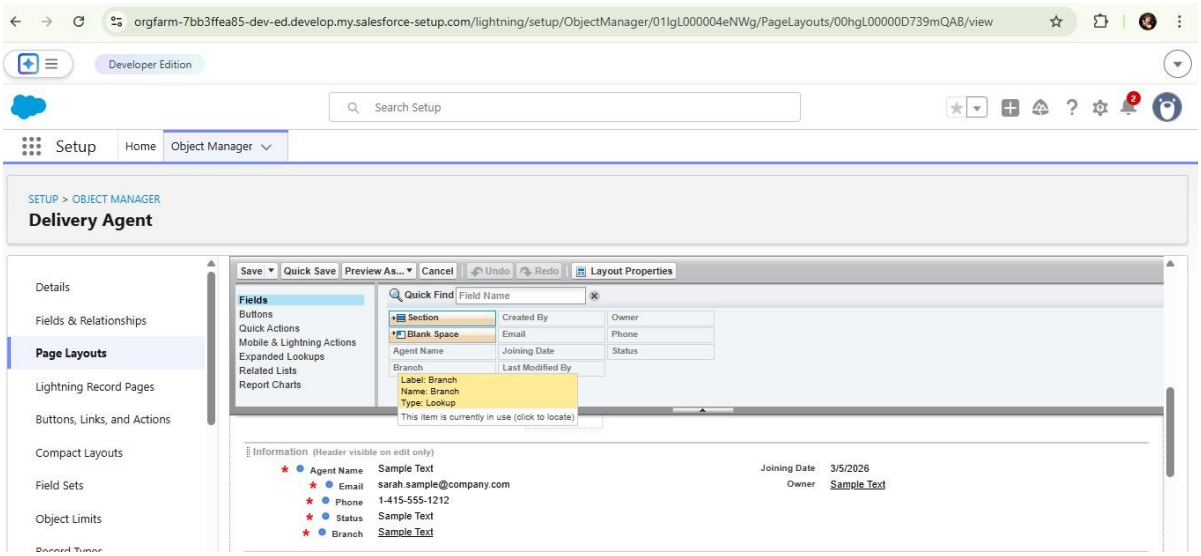
2. It stores details like shipment ID, date, status, and delivery address.

3. Shipment helps companies monitor orders during transportation.

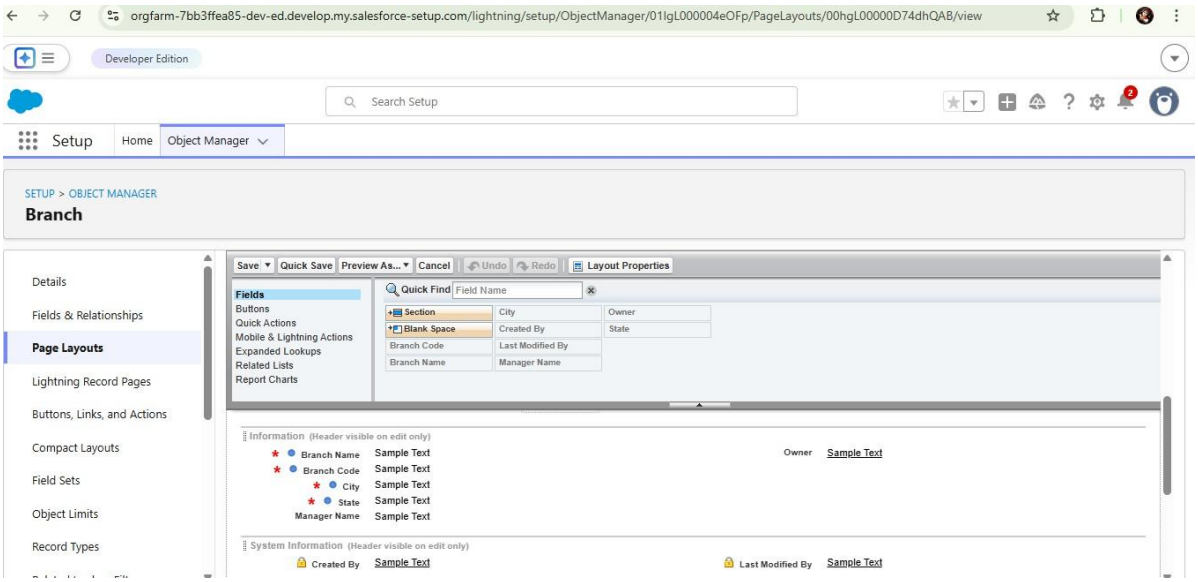
4. It improves order management and delivery tracking in the system.



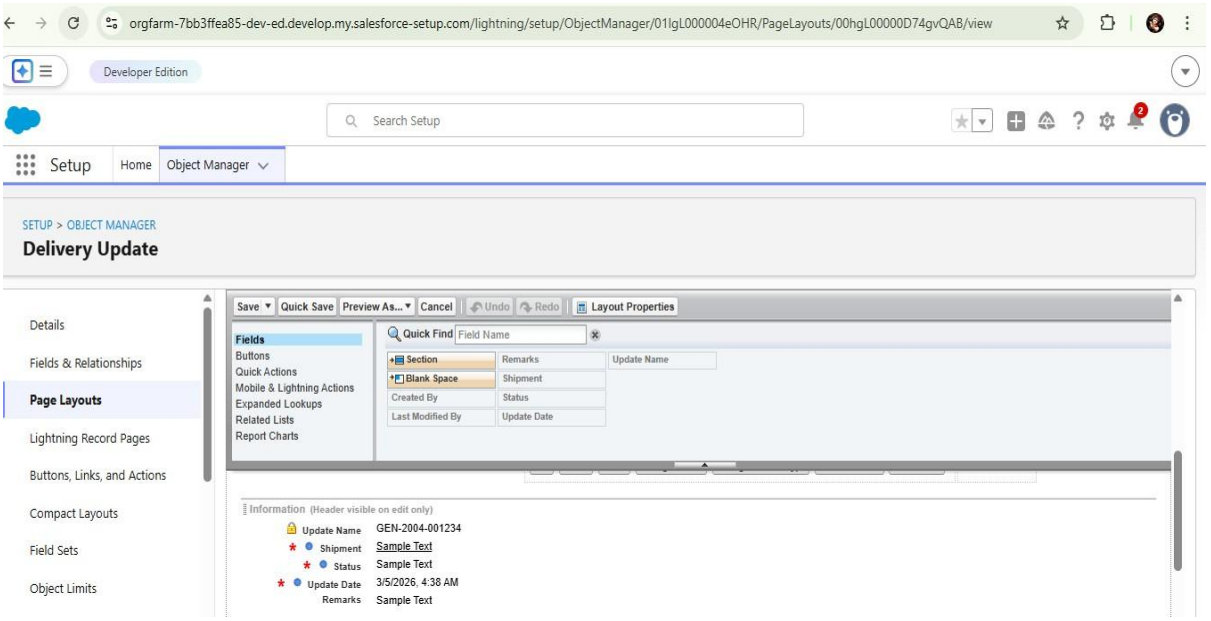
Page Layout for Delivery Agent



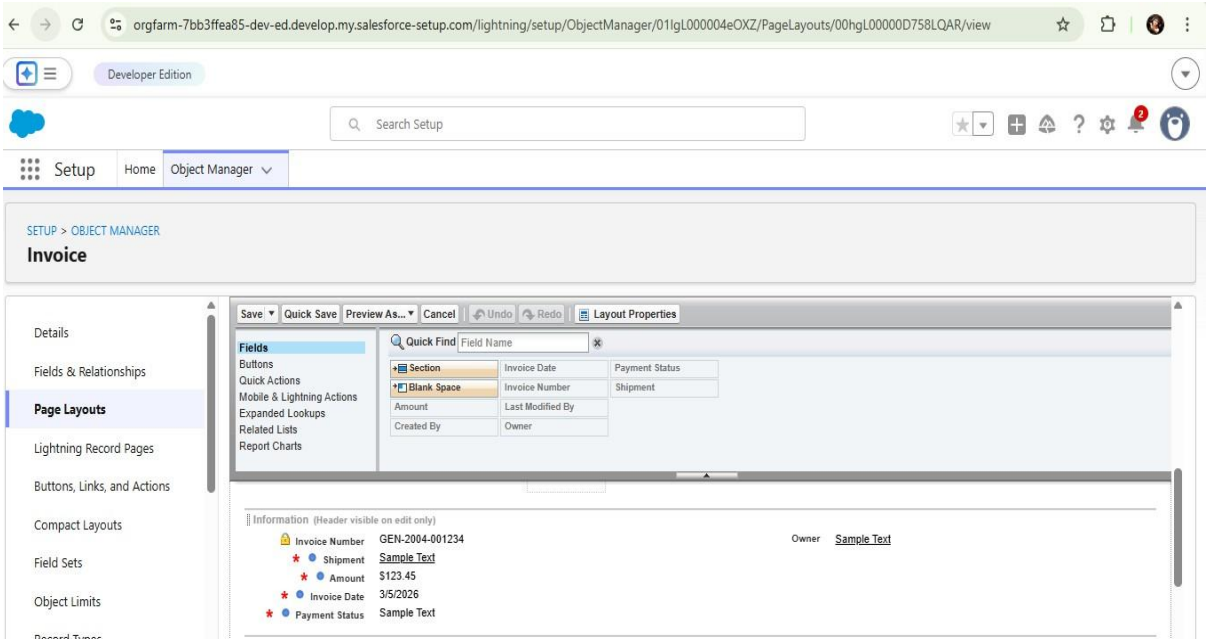
Page Layout for Branch



Page Layout for Delivery Update

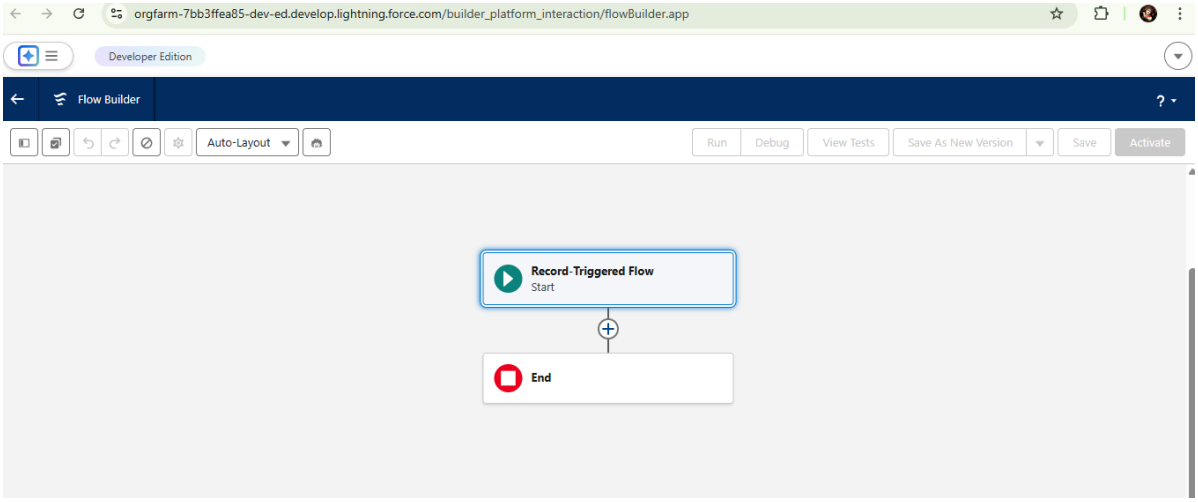


Page Layout for Invoice



Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

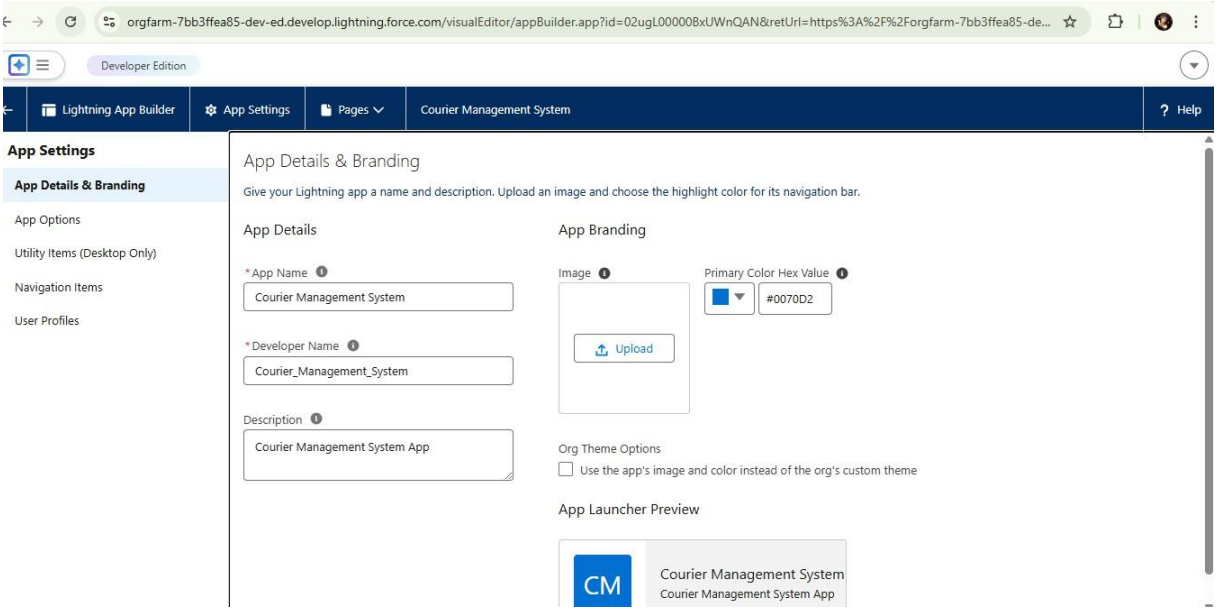


The Lightning App :

What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Create a Lightning App



Profile:

What is a profile?

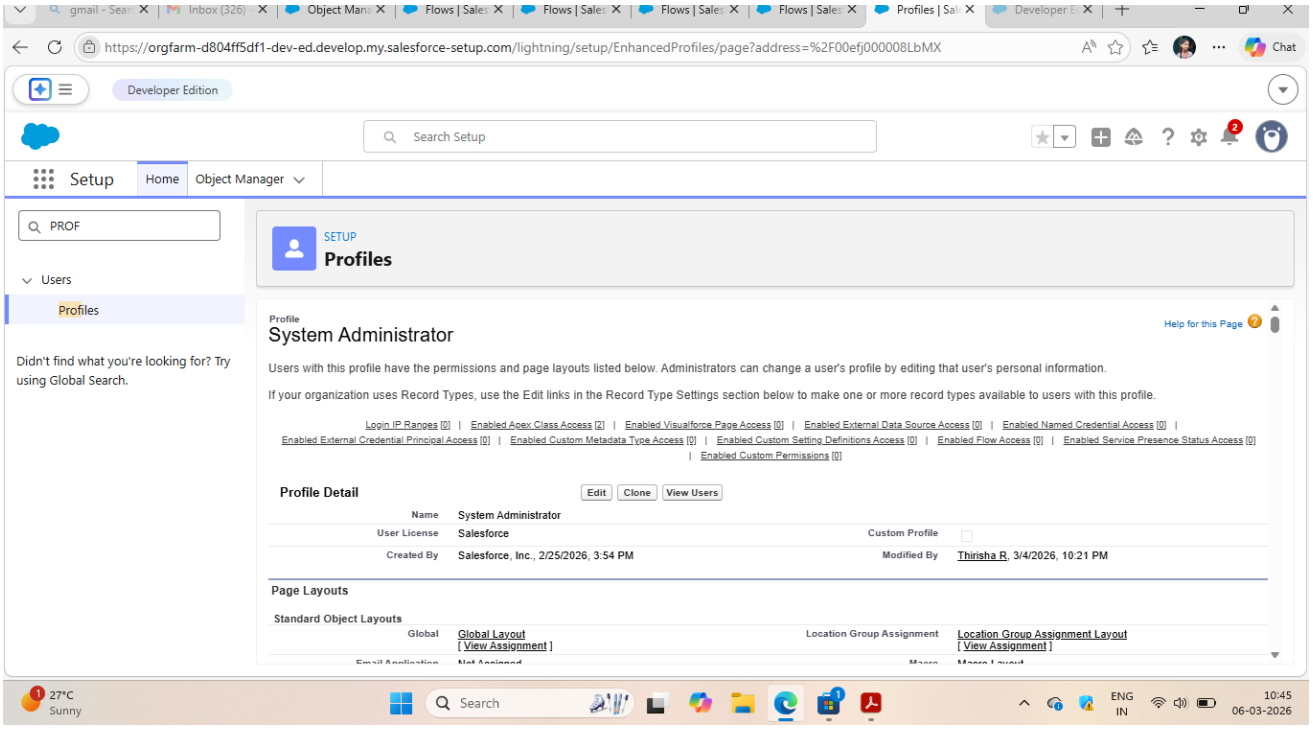
A profile is a group/collection of settings and permissions that define what a user can do in salesforce.

A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types.

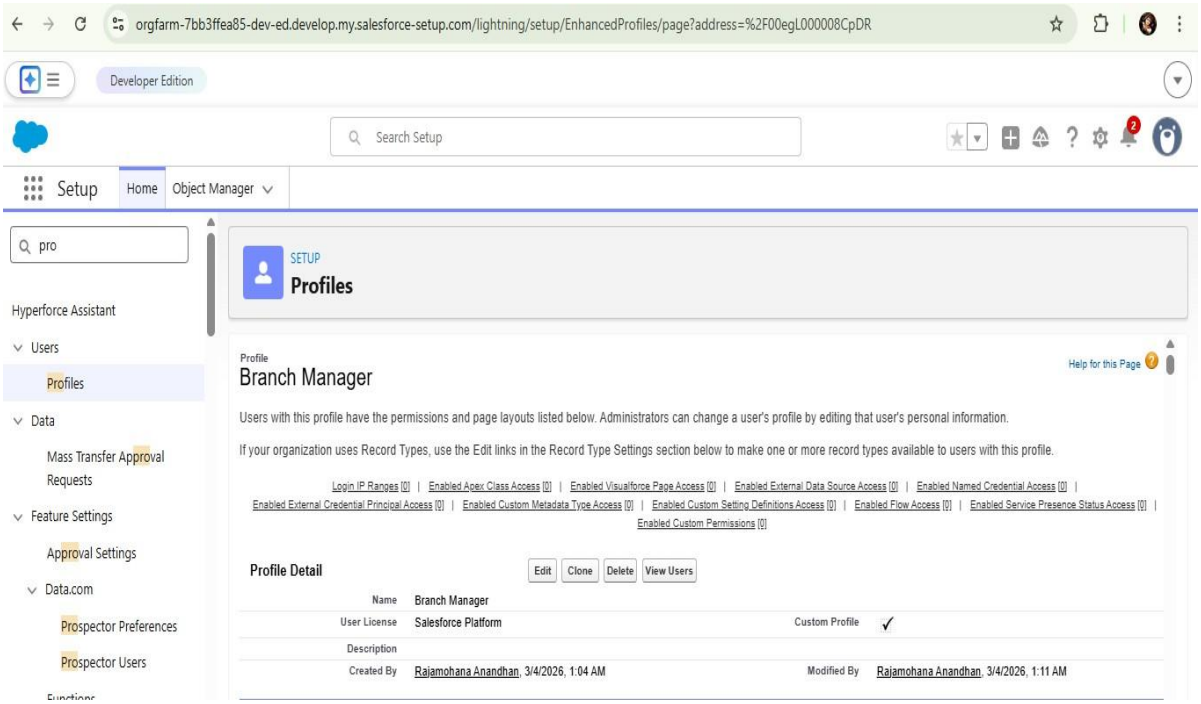
Login hours & Login IP ranges**.**

1. Profile controls what users can see and do in Salesforce.
2. It defines permissions like create, read, edit, and delete for records.
3. Profiles also control access to objects, tabs, and apps.
4. Every Salesforce user must have one profile assigned.
5. Profiles help maintain security and manage user access in the system.

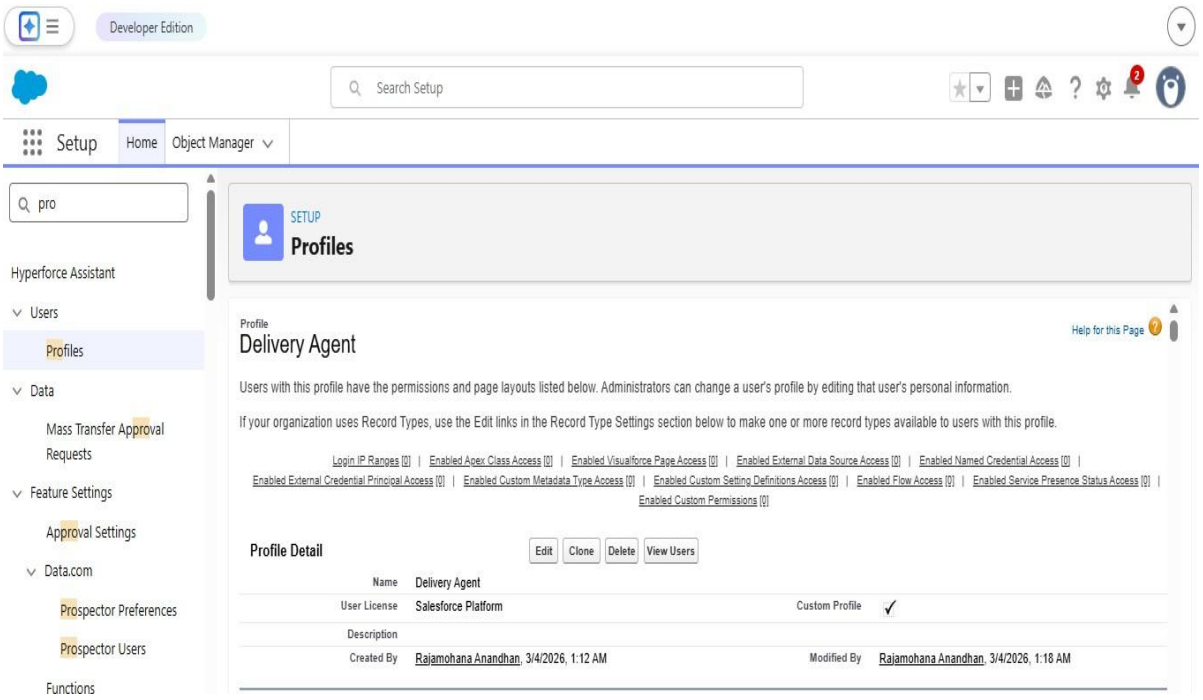
Courier Admin



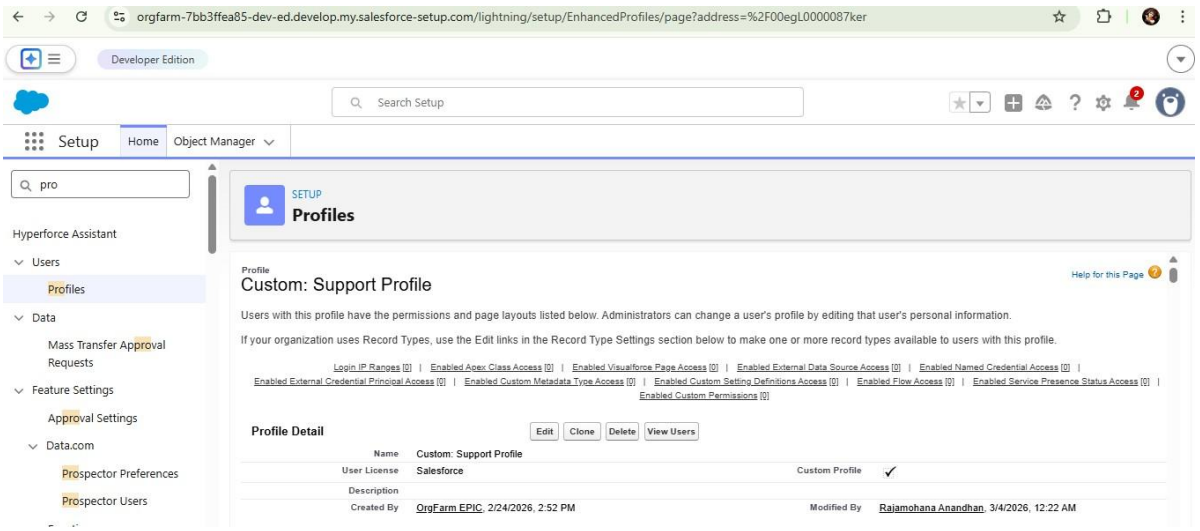
Branch Manager



Profile – Delivery Agent



Profile – Customer Support

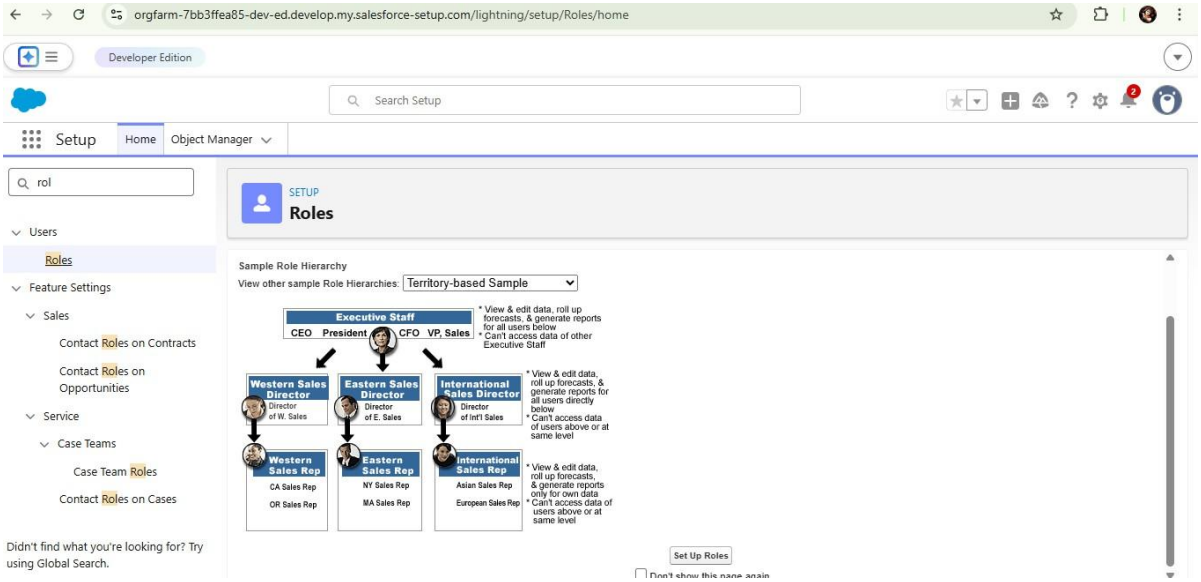


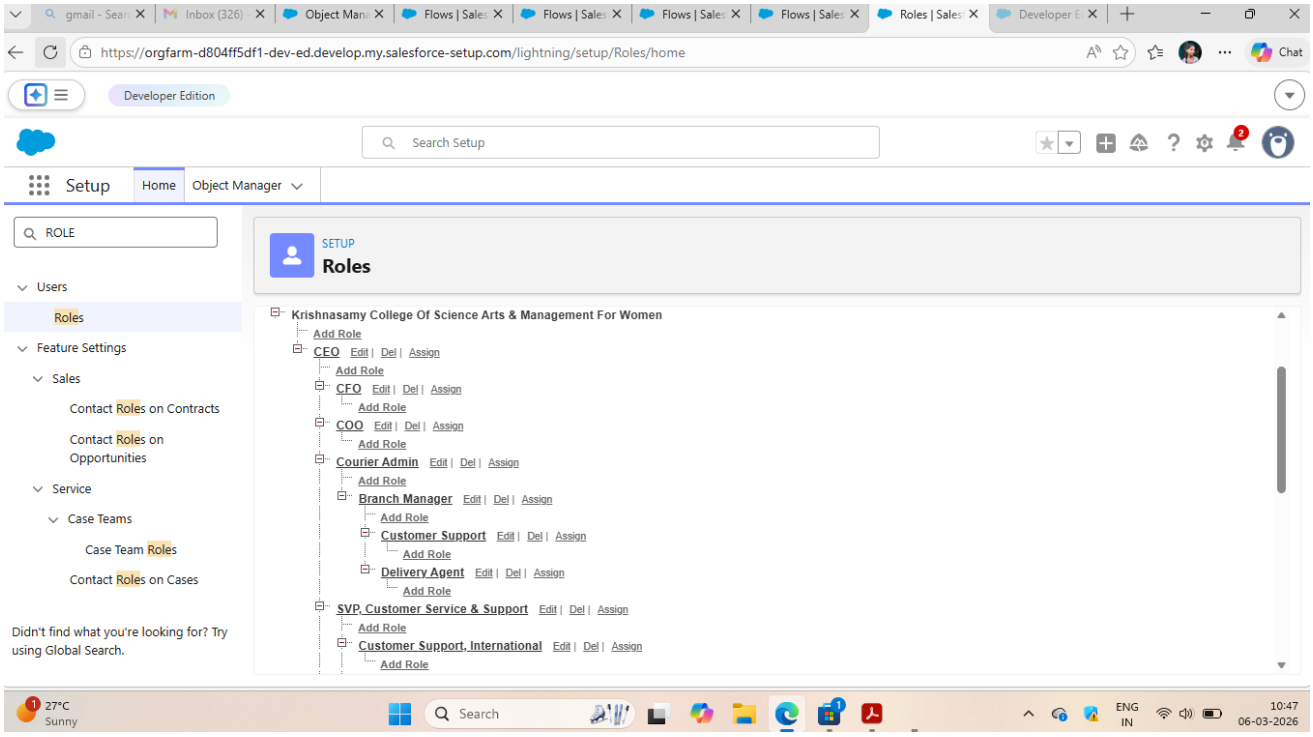
Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system.

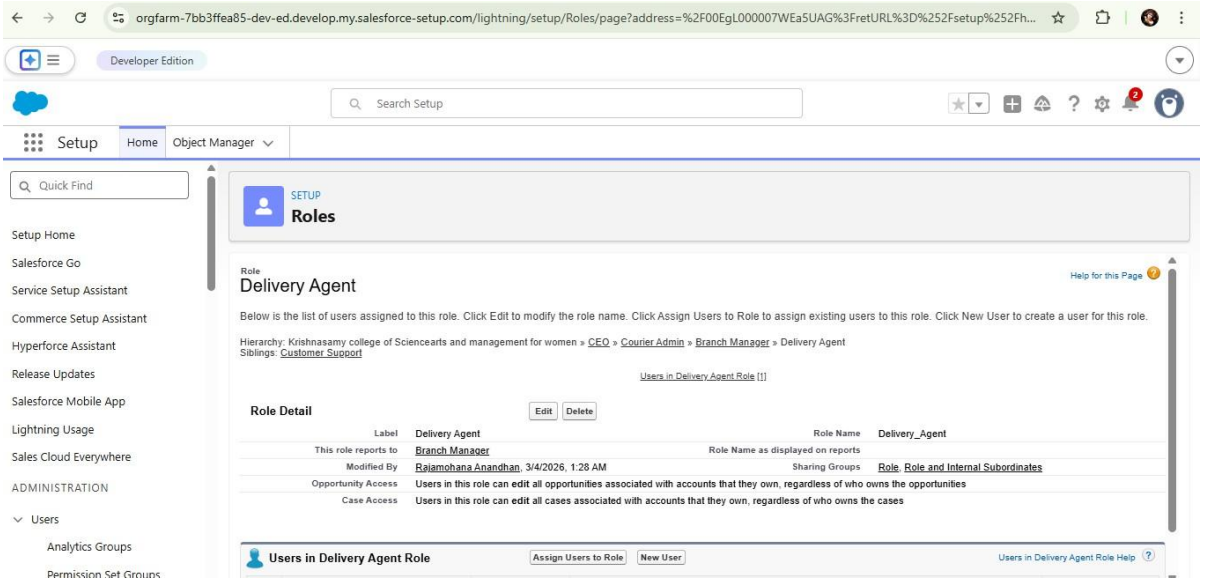
They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Creation of Courier Admin Role

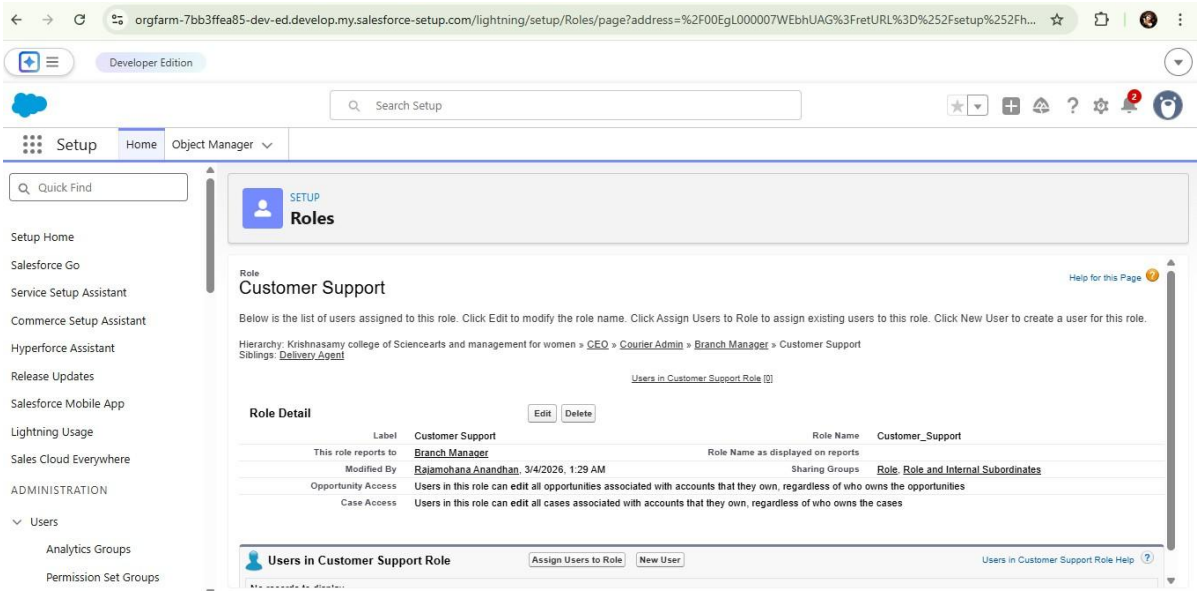




Delivery Agent



Customer support in role



User :

What is a user?

A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records.

Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

In Salesforce:
User (5 lines)

1. A **User** is a person who can **log in and use Salesforce**.
2. Each user has **login details, username, and password**.
3. Users are assigned a **profile to control their permissions**.
4. They can **create, view, edit, or manage records** based on access.
5. Users represent **employees who work in the Salesforce system**.

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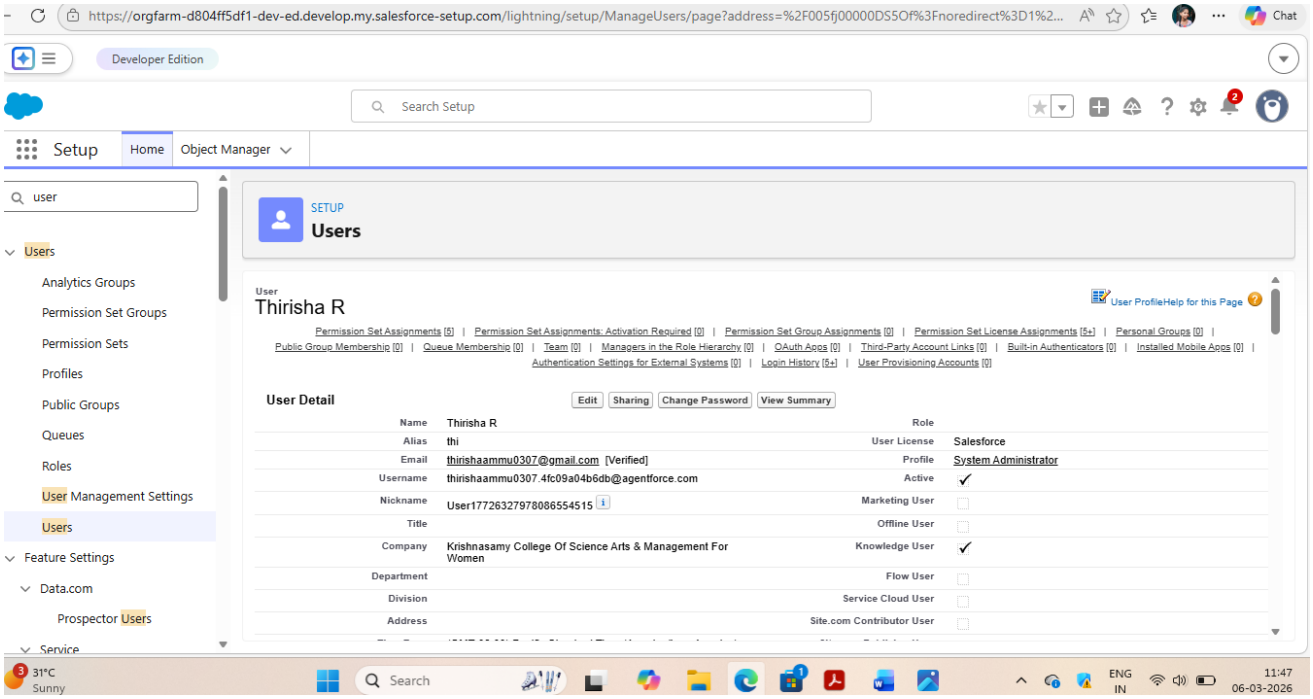
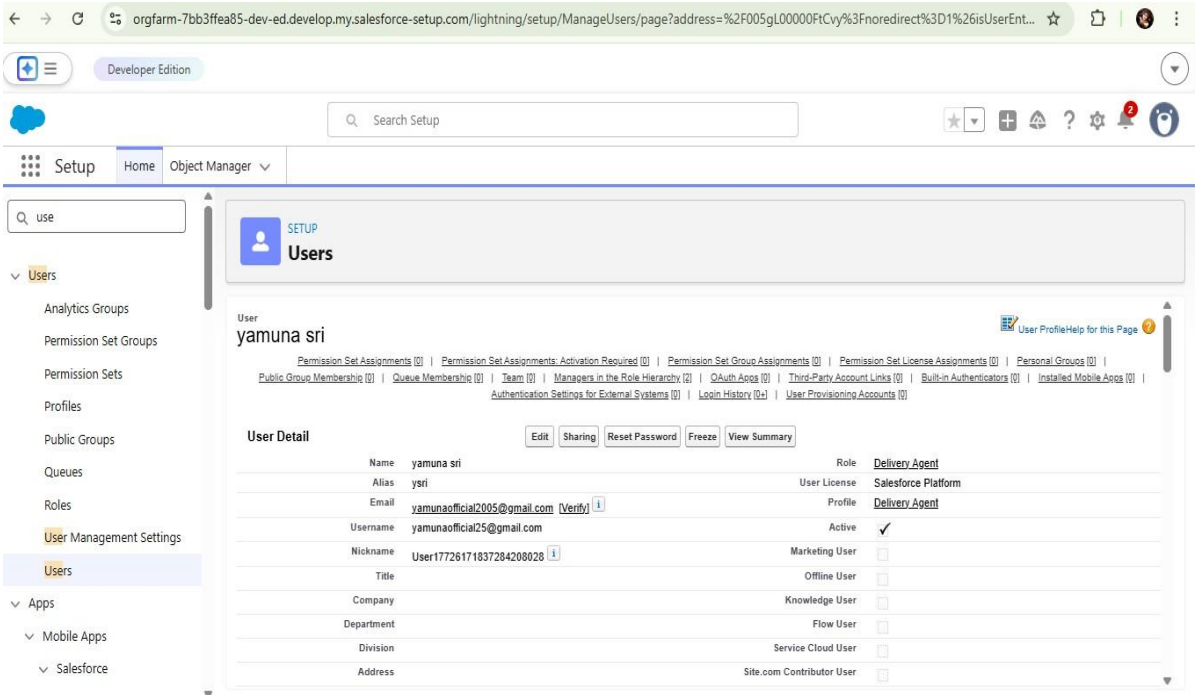
To Create a User

User details





Reports and Dashboards

In Salesforce, Reports and Dashboards are tools used to analyze and display data stored in the system. Reports help users collect, organize, and summarize records based on specific criteria.

Dashboards present this report data in a visual format such as charts, graphs, and tables. They provide a quick overview of important business information. Reports and dashboards help organizations monitor performance and make better decisions.

1. **Reports** are tools used to **organize and summarize data stored in Salesforce**.

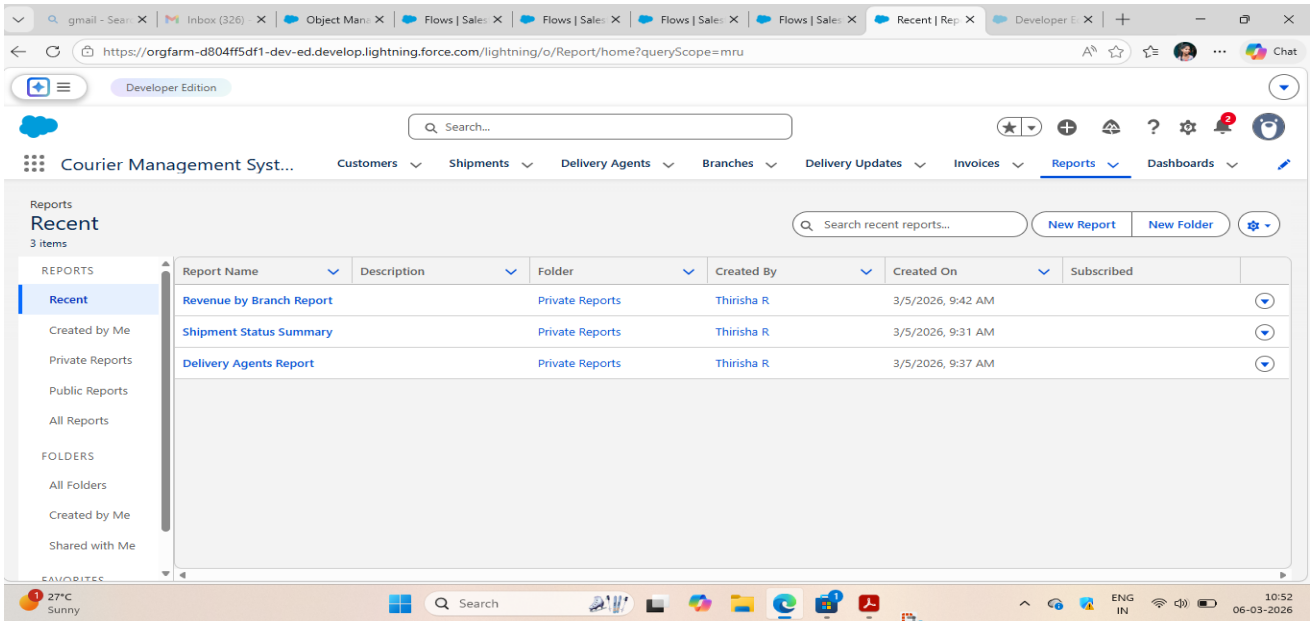
2. They allow users to **filter, group, and analyze records easily**.

3. **Dashboards** display report data in **visual components like charts and tables**.

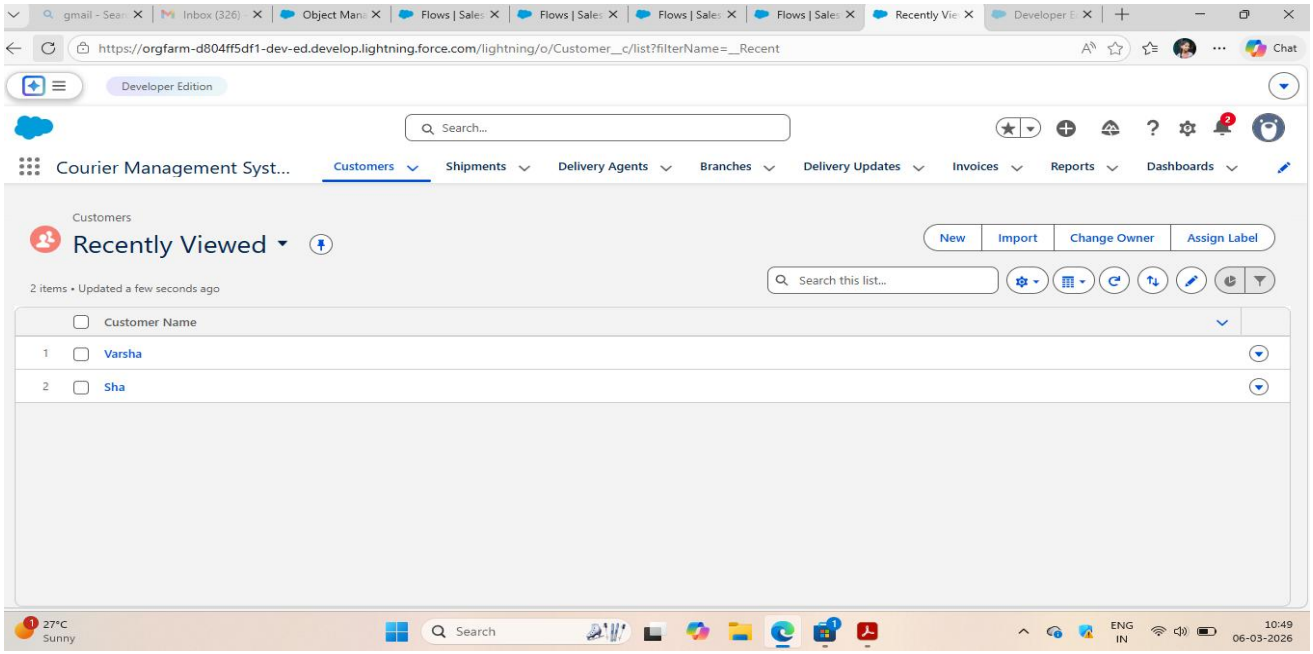
4. They provide a **real-time overview of business performance**.

5. Reports and dashboards help managers **monitor activities and make better decisions**.

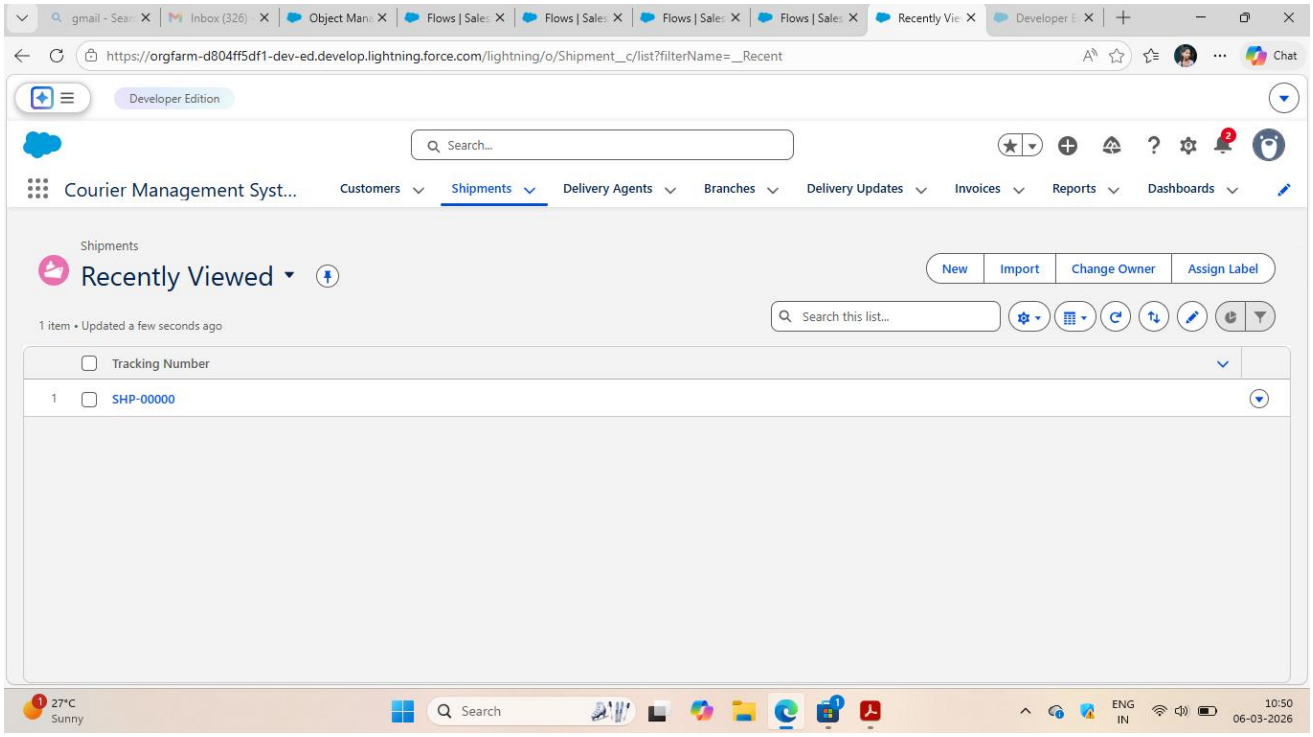
Reports



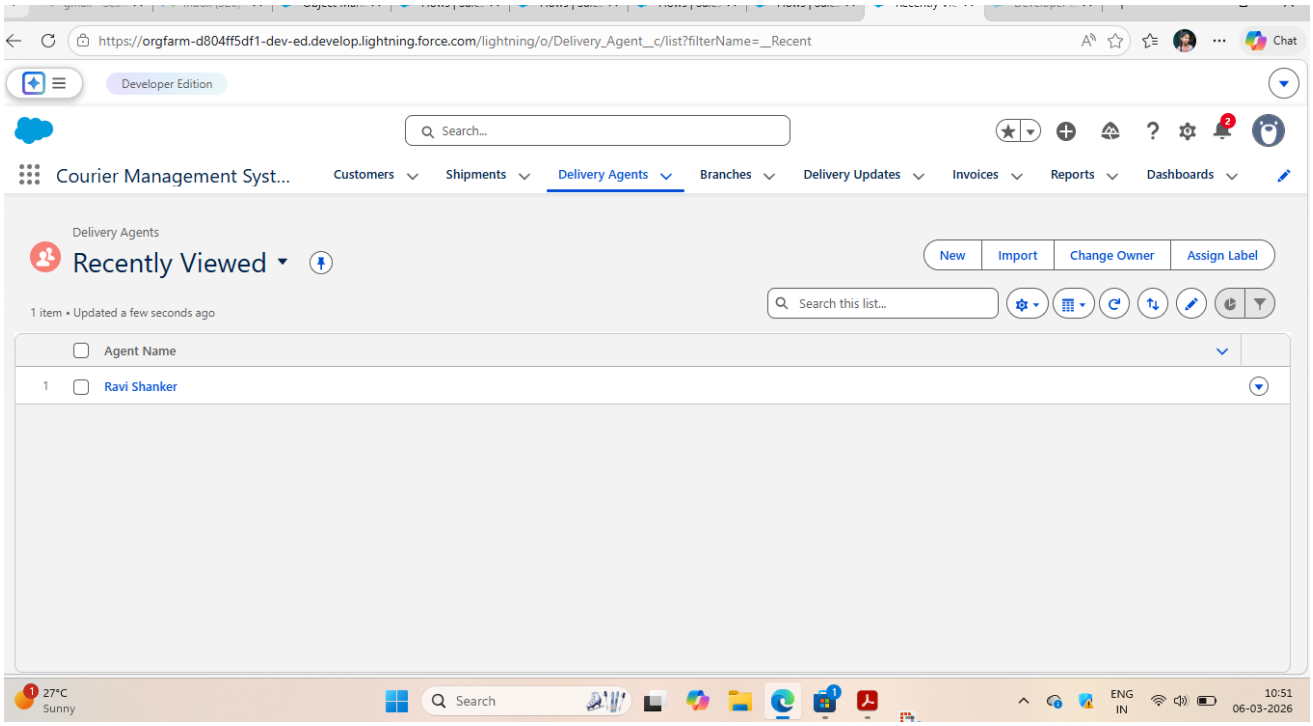
Customer



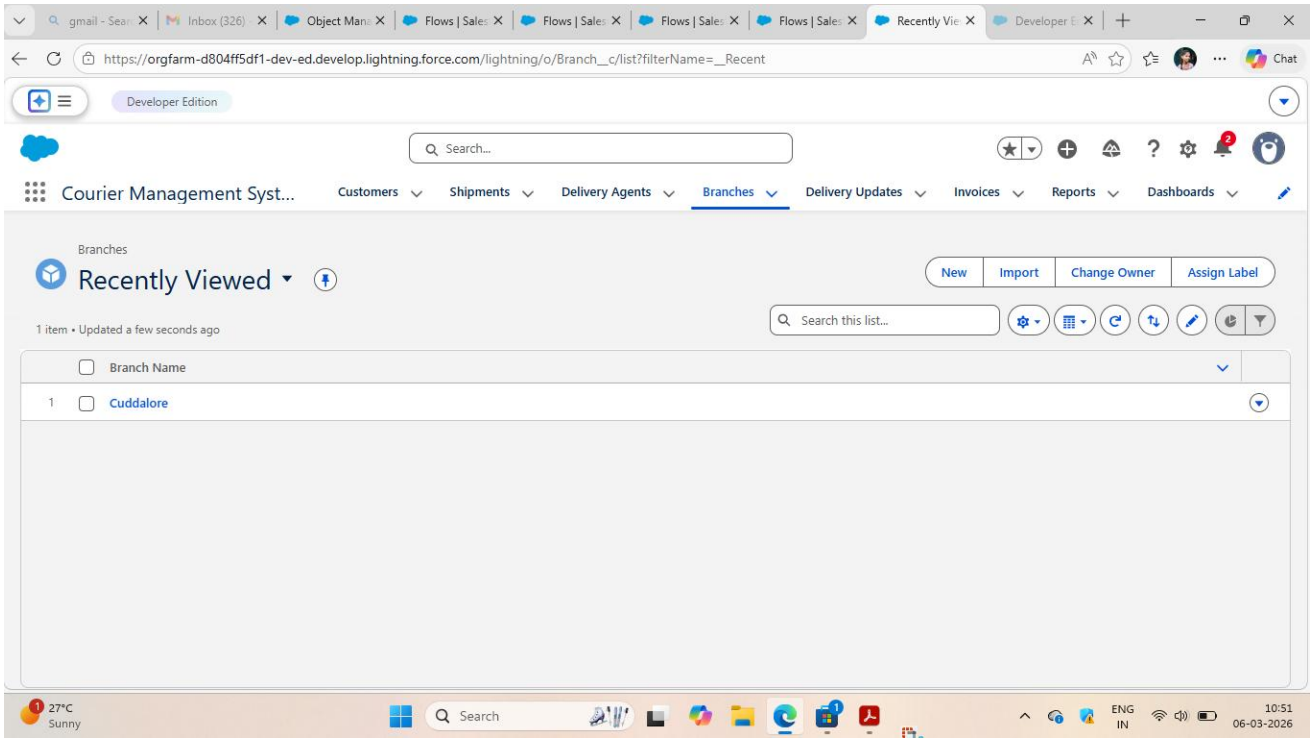
Shipment



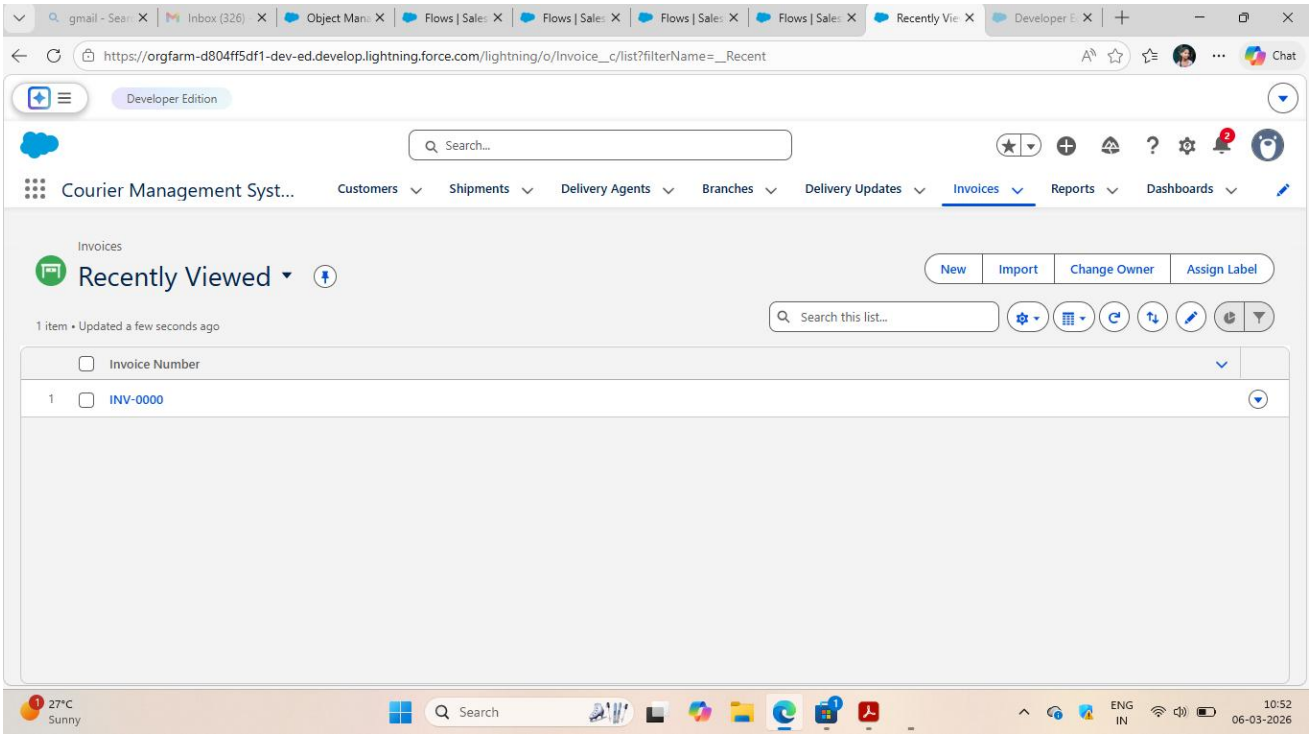
Delivery Agent



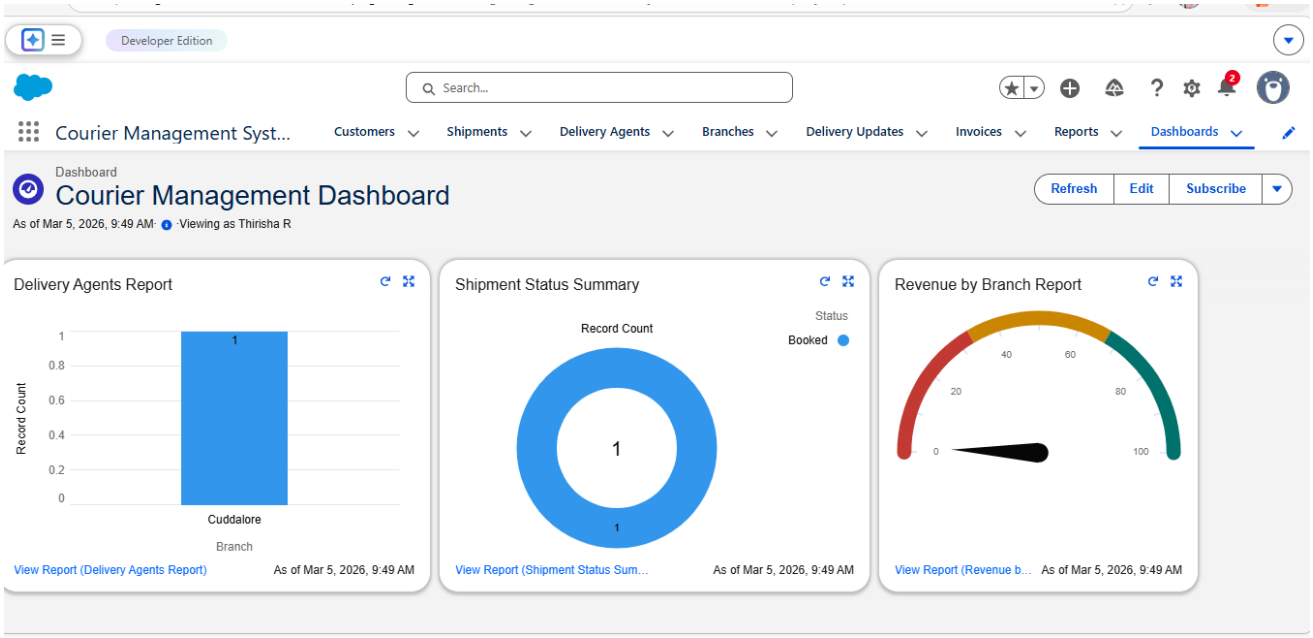
Branch



Invoice



Final Layout



FUNCTIONAL & PERFORMANCE TESTING

Functional Testing

Functional testing is a type of software testing that checks whether the **software functions or features work according to the requirements**. It verifies the system by testing inputs and checking the expected outputs. This testing

focuses on **what the system does** rather than how it works internally.

Functional testing ensures that every feature like login, registration, and data processing works correctly. It helps confirm that the software meets the **user and business requirements**.

Performance Testing

Performance testing is used to check how well a software system **works under different conditions such as speed, load, and stability**. It measures the **response time, scalability, and reliability** of the application. This testing helps identify problems when many users access the system at the same time. Performance testing ensures that the system works **efficiently without slowing down or crashing**. It improves the **overall performance and user experience** of the application.

Importance of Courier Management System

- Helps to track parcels and shipments easily.
- Reduces manual work and human errors.
- Improves delivery speed and efficiency.
- Helps in managing customer and delivery records.
- Provides better communication and tracking updates to customers.
- Makes the courier process more organized and systematic.
- Saves time and operational cost for the company.

ADVANTAGES & DISADVANTAGES

Advantages of Courier Management System

1. It helps to track shipments easily and accurately.
2. It reduces manual work and saves time.
3. It improves delivery speed and efficiency.
4. Customers can check the status of their parcels online.
5. It helps companies manage orders, records, and deliveries systematically.

Disadvantages of Courier Management System

1. The system may have high setup and maintenance cost.
2. It depends heavily on internet and technology.
3. System errors or technical issues can affect operations.

4. Employees may need training to use the system properly.

5. There may be security risks if data is not protected properly.

Conclusion:

The Courier Management System helps courier companies manage shipments, deliveries, and customer information efficiently. It reduces manual work and improves the speed and accuracy of parcel tracking.

The system also helps customers check delivery status easily. Even though it may require technology and maintenance cost, it makes the courier process faster, organized, and more reliable. Overall, it improves the quality of courier services and customer satisfaction.

FUTURE SCOPE

The Courier Management System can be improved in the future by adding advanced technologies and features.

It can include real-time GPS tracking to monitor shipments more accurately. Mobile applications can be developed so customers can track parcels easily from their phones.

The system can also use AI and automation to improve delivery planning and efficiency.

In the future, it will help courier companies provide faster, smarter, and more reliable delivery services.

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org